CSDS 611 – Advanced topics and methods of Data Science I

Final project

Due: Tuesday, December 1, by 8:00PM

Problem description: The Lotka-Volterra model is a set of mathematical equations that describes how prey x and predator y populations interact and change over time t in an ecosystem. The model consists of an initial value problem

$$\begin{aligned} t &\in [t_{\mathsf{min}}, t_{\mathsf{max}}] \\ x(t_{\mathsf{min}}) &= \eta_1 \\ y(t_{\mathsf{min}}) &= \eta_2 \\ \frac{dx}{dt} &= +\theta_1 x - \theta_2 xy \\ \frac{dy}{dt} &= -\theta_3 y + \theta_4 xy \end{aligned}$$

In this project, we consider the Lotka-Volterra model with

$$\begin{aligned} t_{\mathsf{min}} &= 0 \\ t_{\mathsf{max}} &= 3 \times 10^6 \\ \eta_1 &= 2.11 \\ \eta_2 &= 1.35 \end{aligned}$$

and the provided data of measurements $x(t_n), y(t_n)$ at times t_n, and seek to estimate the values of the parameters $\theta_{1:4}$ within the following parameter ranges

$$\begin{aligned} \theta_1 &\in [0.10, 50.0] \\ \theta_2 &\in [0.01, 2.00] \\ \theta_3 &\in [0.10, 5.00] \\ \theta_4 &\in [0.01, 2.00] \end{aligned}$$

Step 1: Cast the estimation of $\theta_{1:4}$ as a least-squares optimization problem. Implement a Runge-Kutta numerical integrator to evaluate the solution of the initial value problem and the objective. Benchmark the execution time of a single objective evaluation.